

YIBO PENG

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EDUCATION

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering

Pittsburgh, PA

Aug 2023 – May 2025

Beijing Jiaotong University & Lancaster University

Bachelor of Science in Computer Science (Honours)

Beijing, CN & Lancaster, UK

Aug 2018 – July 2022

PUBLICATIONS

- [1] **Y. Peng**, S. Chen, L. Lian, D. Wagner. “[SecOPD: Mitigating Adaptive Prompt Injections by On-Policy Distillation.](#)” in Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2026 Main**), Budapest, Hungary, October 2026.
- [2] P. Xia*, **Y. Peng***, J. Wang*, K. Zeng, X. Wu, X. Tang, H. Zhu, Y. Li, S. Liu, Y. Lu, H. Yao. “[MMedAgent-RL: Optimizing Multi-Agent Collaboration for Multimodal Medical Reasoning.](#)” in Proceedings of the 14th International Conference on Learning Representations (**ICLR 2026**), Rio de Janeiro, Brazil, April 2026. [**Paper**]
- [3] **Y. Peng***, J. Song*, L. Li*, X. Yang, M. Christodorescu, R. Mangal, C. Pasareanu, H. Zheng, B. Chen. “[When *Correct* Is Not Safe: Can We Trust Functionally Correct Patches Generated by Code Agents?](#)” in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026 Main**), San Diego, California, July 2026. [**Paper**]
- [4] **Y. Peng***, P. Xia*, D. Zhong*, K. Zeng, S. Han, Y. Zhou, J. Liu, R. Zhang, H. Yao. “[SimpleOCR: Rendering Visualized Questions to Teach MLLMs to Read.](#)” in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL 2026 Findings**), San Diego, California, July 2026. [**Paper**]
- [5] **Y. Peng**, Z. Wang, D. Fried. “[Can Long-Context Language Models Solve Repository-Level Code Generation?](#)” (**ITI Student Research Symposium**), 2025 (Poster). [**Paper**]
- [6] S. Chen, **Y. Peng**, C. Sitawarin, D. Wagner. “[SecPO: Principled Adversarial Training for Prompt Injection Security.](#)” *Submitted to the 36th USENIX Security Symposium* (**USENIX Security 2027**).
- [7] M. N. Le*, N. Gondi*, **Y. Peng**, R. Ni, L. Jia, B. Chen, H. Zheng. “[Self-Evolving Defense: Continual Security Policy Learning for LLM Agents.](#)” *Submitted to the 2027 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics* (**NAACL 2027**).

RESEARCH EXPERIENCE

Berkeley Security Group, University of California, Berkeley

Researcher

Feb 2026 – Present

- Researched prompt-injection security for LLMs and AI agents
- Developed model-level defenses using post-training and preference optimization
- Evaluated security-utility tradeoffs under prompt-injection attacks across instruction-following and agentic tool-calling settings
- Advisor: [David Wagner](#)

InfiniAI Lab, Carnegie Mellon University

Researcher

May 2025 – Present

- Researched security and reliability risks in LLM-based code agents
- Researched vulnerability injection in functionally correct code patches
- Built evaluation pipelines for repository-level code repair and code-agent robustness
- Advisor: [Beidi Chen](#)

- Researched multimodal language models and visual text understanding
- Researched OCR-oriented training methods for improving visual grounding
- Evaluated multimodal reasoning models on OCR-intensive benchmarks
- Advisor: [Huaxiu Yao](#)

- Researched multi-agent systems for multimodal medical reasoning
- Researched reinforcement learning methods for coordinating medical reasoning agents
- Evaluated multimodal medical agents on visual question answering benchmarks
- Advisor: [Yan Lu & Huaxiu Yao](#)

- Researched code LLMs for repository-level code generation and code understanding
- Researched retrieval-augmented generation and long-context modeling for coding tasks
- Built RAG-based semantic code search tools for LLM code agents
- Advisor: [Daniel Fried & Graham Neubig](#)

WORK EXPERIENCE

PricewaterhouseCoopers LLP (PwC)

Beijing, CN

Development Engineer Intern – Quantitative Model Expert Team

Nov 2021 – Apr 2022

- Developed a large VBA application to assess and calculate Expected Credit Loss (ECL) of accounts receivable.
- Reduced calculation time from **15 minutes to 10 seconds** by transitioning calculations to the database.
- Improved code efficiency by simplifying loops, reducing global variable usage, and optimizing function calls.
- Collaborated with cross-functional teams to integrate the model and over **230 listed companies** used it.

SERVICES

Reviewer: ACM Computing Surveys, NeurIPS 2026, CVPR 2026, NeurIPS 2025 Efficient Reasoning Workshop; NeurIPS 2025 ResponsibleFM Workshop; ICML 2025 R2-FM Workshop